

2. Hello World!

A *"Hello World!"* in the Arduino sphere is a blinking LED.

You just need an Arduino board and a USB cable.

pen a new file in the IDE. The lines of code below are pre-written and form the basis of every program. We'll explain more about that later.

```
void setup() {  
  // put your setup code here, to run once:  
  
}  
  
void loop() {  
  // put your main code here, to run repeatedly:  
  
}
```

Give the file a name and save it.

Then type the following text into the Arduino sketch editor, but you can skip the lines starting with a `//` as they are comments.

```
// LED connected to digital pin 13 > works for UNO & Every  
const int ledPin = 13;  
  
// the setup function runs once when you press reset  
// or power the board  
void setup() {  
  // initialize digital pin 13 as an output.  
  pinMode(ledPin, OUTPUT);  
  
}  
  
// the loop function runs over and over  
void loop() {  
  // turn the LED on (HIGH is the voltage level)  
  digitalWrite(ledPin, HIGH);  
  // wait for 1000 milliseconds or 1 second  
  delay(1000);  
  // turn the LED off by making the voltage LOW  
  digitalWrite(ledPin, LOW);  
  // wait for another second  
  delay(1000);  
}
```

Click the **Verify** button to check whether your code is correct.

If everything is fine, you'll see the message **"Done compiling"** appear at the bottom of the Arduino IDE. The Arduino IDE has translated your sketch into an executable program that can be run by the board.

Now, click the **Upload** button. This will reset the board and stop its current functions. The compiled sketch is then sent to the board and stored in its memory. Afterwards, the board will run it. Subsequently the board will run it.

If everything went well you'll see the notifications **"Done compiling."** and **"Done uploading."** appear to let you know the process has completed correctly.

You can adjust the two delay values to change the blinking rhythm. Don't forget to compile and upload the code after making changes.